



Dapagliflozin administration for 1 year promoted kidney enlargement in patient with ADPKD

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Abstract

To date, there is insufficient evidence regarding use of sodium–glucose cotransporter-2 (SGLT2) inhibitors for patients with autosomal-dominant polycystic kidney disease (ADPKD), as such cases have been excluded from previous clinical trials exploring the kidney protection effects of such medications. Here, findings of an ADPKD patient who received dapagliflozin, a selective SGLT2 inhibitor, for 1 year are presented. A 38-year-old woman with a family history of ADPKD wished for treatment with dapagliflozin. After starting administration at 10 mg/day, total kidney volume (TKV) continued to increase, from 1641 to 1764 mL after 84 days and then to 2297 mL after 340 days. The estimated glomerular filtration rate (eGFR) was also decreased from 67.3 to 56.2 mL/min/1.73 m², and then to 51.4 mL/min/1.73 m² at those times. Immediately after discontinuation of dapagliflozin, TKV and eGFR were slightly improved to 2263 mL and 55.1 mL/min/1.73 m², respectively. Following a review of basic research studies, we consider that increased intratubular urinary osmotic pressure, compensatory glucose reabsorption by sodium–glucose cotransporter-1 in the late proximal tubule, and hypertrophy shown in collected cells caused by increased vasopressin may be associated with ADPKD disease progression. Caution may be needed when administering dapagliflozin to patients with ADPKD.

Keywords Autosomal-dominant polycystic kidney disease · Sodium-glucose cotransporter-2 inhibitors · Total kidney volume

Introduction

Autosomal-dominant polycystic kidney disease (ADPKD) is an inherited disorder characterized by development of fluid-filled cysts in the kidneys [1]. As the cysts grow, the kidneys progressively become enlarged and kidney function deteriorates. ADPKD is one of the causes of chronic kidney disease (CKD) and the fourth leading cause of end-stage

kidney disease (ESKD), which requires kidney replacement therapy such as hemodialysis [2]

Glucose metabolism impairment has been highlighted as a key feature and important disease modulator in ADPKD cases [3]. Cyst epithelial cells are dependent on aerobic glycolysis, while hyperglycemia promotes cystogenesis and cyst growth [4]. Mild-to-moderate food restriction has been shown to delay cystogenesis and kidney function deterioration in a progressive mouse model [5] and also an orthologous mouse model [6] of ADPKD. Furthermore, rats with polycystic disease (PKD) derived from ketosis and treated with β -hydroxybutyrate showed reduced cyst growth as compared with those on a control diet [7]. In ADPKD patients, overweight and obesity are factors associated with an increase in total kidney volume (TKV) [8], while daily calorie restriction and intermittent fasting resulting in decreased body weight were found to slow the increase in TKV in a cohort of overweight or obese ADPKD patients [9].

For inhibiting glucose reabsorption by the renal proximal tubules, administration of a sodium–glucose cotransporter-2 (SGLT2) inhibitor may result in

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reduced cystogenesis. Recently, the presented results of the DAPA–CKD trial [10] and EMPA–Kidney trial [11] indicate kidney protective effects of such medications in CKD patients both with and without diabetes. However, ADPKD patients were excluded from both of those randomized control trials, thus evidence regarding the use of SGLT2 inhibitors for such cases remains insufficient. Furthermore, in our recent retrospective pilot study regarding the short-term effects of dapagliflozin, a selective SGLT2 inhibitor administered for ADPKD, the average observational period of the patients was short at 102 days [12]. Thus, the long-term effects of that medication remain largely unknown.

The present case report describes an ADPKD patient given the SGLT2 inhibitor dapagliflozin for 1 year, during which TKV was determined with a SYNAPSE VINCENT® high-speed 3D-image analysis system (Fuji Medical Systems, Tokyo, Japan). After starting administration, the TKV was remarkably increased, though immediately decreased after discontinuation. Findings in the present case suggest unfavorable effects of dapagliflozin leading to kidney enlargement as well as kidney function deterioration in ADPKD patients, unlike other etiological factors of CKD, such as diabetic kidney disease and glomerulonephritis.

Case report

A 38-year-old woman was diagnosed with ADPKD based on results of abdominal ultrasound screening performed for abdominal pain at the age of 16 years. A family history of the disease was noted, including her mother who had CKD, as well as her uncle, grandmother, and great-grandmother who were affected by ESKD requiring hemodialysis. The patient was found to have multiple liver cysts, though no other complications were noted, such as brain aneurysm or heart valve disease. An abdominal computer tomography (CT) examination performed on April 2017 revealed enlargement of the bilateral kidneys with multiple cysts of various sizes and a TKV of 1417 mL (Fig. 1a). The patient was referred to our hospital on April 2018 due to retirement of the attending physician. As for disease progression, she was given the Mayo imaging classification of 1D [13]. Treatment with tolvaptan, a vasopressin V₂ receptor antagonist, was recommended after providing comprehensive information regarding its effects and the clinical course of ADPKD. However, the patient declined that medication due to concerns regarding possible effects of its aquaretic side-effect on her work as well as kidney function deterioration noted in her mother despite receiving such treatment. Hypertension was noted along with treatment with antihypertensives including olmesartan (10 mg/day). One year later, her mother was diagnosed with ESKD and required

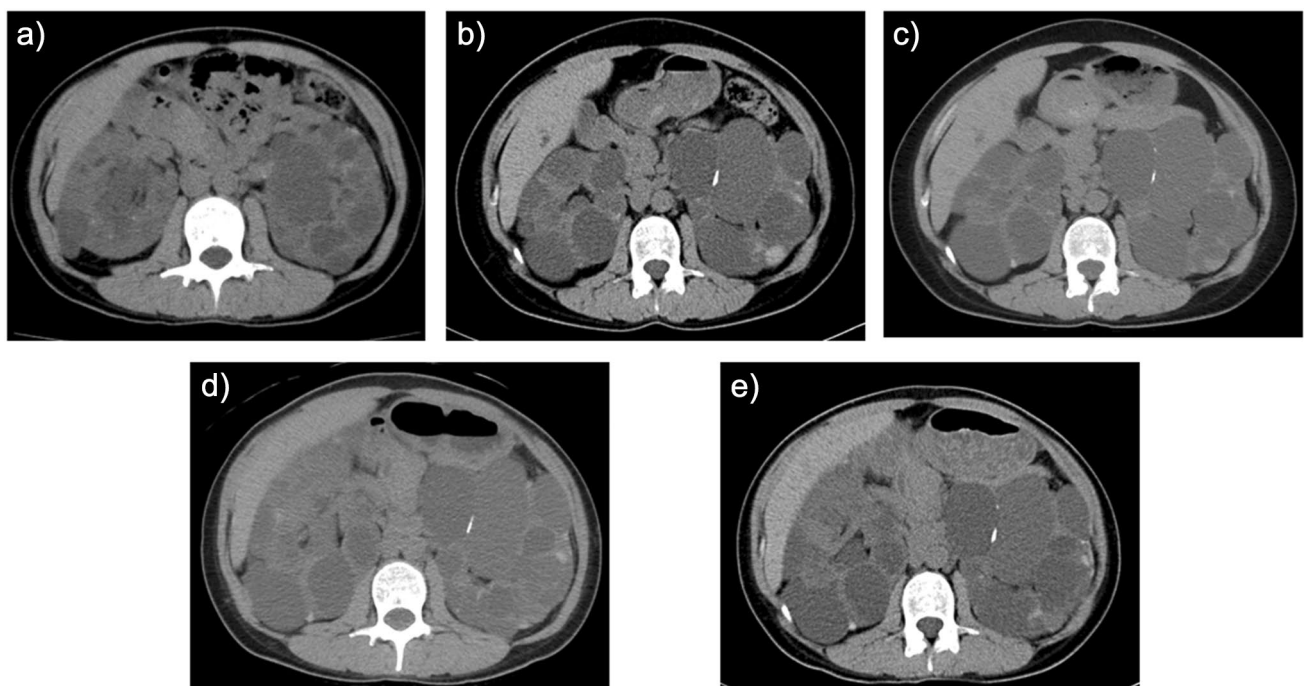


Fig. 1 Computer tomography images of abdominal area. Dapagliflozin was administrated from January 2022 to January 2023. **a** April 2017. **b** January 2022. **c** April 2022. **d** January 2023. **e** February 2023

hemodialysis, and the present patient was transferred to the same hospital to be with her mother.

The patient was again transferred to our hospital in April 2020. She was being treated with 20 mg of olmesartan and estimated glomerular filtration rate (eGFR) was 74.6 mL/min/1.73 m². Changes in eGFR and TKV are shown in Fig. 2. On January 2022, eGFR was found to be decreased to 67.3 mL/min/1.73 m² and TKV was increased to 1641 mL (Fig. 1b). The patient had knowledge regarding SGLT2 inhibitors and strongly requested

such treatment, thus dapagliflozin administration at 10 mg/day was started. Blood pressure and laboratory data just before initiation of dapagliflozin are shown in Table 1. A CT examination performed on April 2022 for reapplication for intractable progressive ADPKD revealed a remarkable increase in TKV to 1764 mL (Fig. 1c). Because of the strong family history of ADPKD, initiation of tolvaptan was again recommended, though the patient refused that due to the aquaretic side-effects and for professional

Fig. 2 Clinical course. During administration of dapagliflozin, total kidney volume (TKV) was found to be increased and estimated glomerular filtration rate (eGFR) decreased. Immediately after its discontinuation, both TKV and eGFR showed slight improvements. Solid line: eGFR, gray bar: TKV

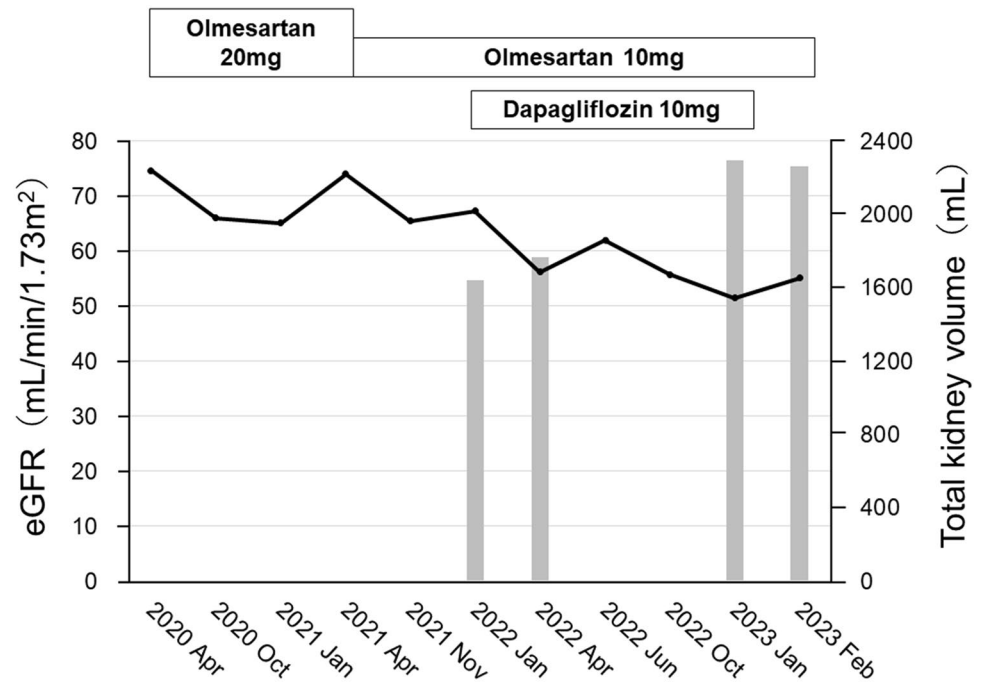


Table 1 Blood pressure and laboratory data

Date	January 2022	April 2022	January 2023	February 2023
Blood pressure (mmHg)	141/106	135/81	120/80	135/85
Creatinine (mg/dL)	0.78	0.92	0.99	0.93
eGFR (mL/min/1.73 m ²)	67.3	56.2	51.4	55.1
Blood urea nitrogen (mg/dL)	20	21	22	21
Sodium (mEq/L)	140	139	140	140
Potassium (mEq/L)	4.0	3.7	4.2	4.5
Chloride (mEq/L)	106	106	106	108
Phosphate (mg/dL)	3.5	3.3	3.3	3.6
Plasma glucose (mg/dL)	NA	90	85	67
HbA1c (%)	NA	5.1	5.2	NA
Triglycerides (mg/dL)	99	113	39	57
Total cholesterol (mg/dL)	230	223	201	214
LDL cholesterol (mg/dL)	118	101	101	96
BNP (pg/mL)	12.0	< 5.8	6.9	NA
Urinary protein (g/gCr)	0.10	0.18	0.21	0.19

BNP brain natriuretic hormone, eGFR estimated glomerular filtration rate, HbA1c hemoglobin A1c, LDL low-density lipoprotein, NA not available

reasons. Although discontinuation of dapagliflozin was also suggested, the patient chose to continue that for 1 year.

During the subsequent dapagliflozin treatment period, TKV continued to increase and was 2297 mL on January 2023 (Fig. 1d), while GFR was decreased to 51.4 mL/min/1.73 m². Finally, the patient agreed to discontinue dapagliflozin and also to initiate tolvaptan treatment. On February 2023, one month after discontinuation of dapagliflozin and the start of hospitalization for initiation of tolvaptan, TKV was slightly improved to 2263 mL and eGFR to 55.1 mL/min/1.73 m² (Fig. 1e). Although changes in body weight during dapagliflozin administration were not noted, that was 49.7 kg at the first visit to our hospital on April 2018 and 48.0 kg at the start of tolvaptan treatment.

Discussion

In the present ADPKD patient, a 1-year administration of dapagliflozin was found to promote kidney enlargement, while increased TKV was immediately improved after its discontinuation. These findings indicate that dapagliflozin treatment accelerated an increase in TKV and also a decrease in eGFR. Although the precise mechanisms by which dapagliflozin promoted disease progression in our patient are unclear, it is important to consider putative factors. It is considered that an increase in intratubular urinary osmotic pressure, compensatory glucose reabsorption by sodium–glucose cotransporter-1 (SGLT1) in the late proximal tubule, and hypertrophy shown in collected cells caused by increased vasopressin were likely associated with disease progression in the present patient.

Previously reported *in vivo* findings showed that dapagliflozin accelerated cystogenesis in PCK rats [14], similar to the present case. It has been speculated that urinary osmotic pressure mediated by glucosuria and compensatory sodium reabsorption associated with dapagliflozin use promote cystogenesis in late proximal tubules, distal tubules, and collecting ducts [14]. In support of that speculation, another study that used a human organoid-on-chip model of PKD found that increased osmotic pressure creates pressure within the intersection and leads to its detachment from neighboring tubules, which undergo deformation and expansion to fill the resultant interstitial space [15]. Furthermore, a greater decrease in urinary osmolality associated with tolvaptan has been shown to result in slower eGFR decline [16]. Tolvaptan was not administered during the time period of the present case study, thus urinary osmolality was not determined. Nevertheless, findings from previous studies indicate that increased urinary osmotic pressure induced by dapagliflozin may increase TKV in a distal part from the late proximal tubule.

Treatment with phlorizin, which inhibits both SGLT1 and SGLT2, resulted in decreased cystic index and kidney to body weight ratio in Ham:SPRD rats [17]. Within the kidneys, SGLT1 is located in the late proximal renal tubule, whereas SGLT2 is found in the early proximal tubule [18]. Under an SGLT2 imbibitional condition, glucose reabsorption by SGLT1 is known to occur as a compensatory increase [19]. Absorbed glucose in the proximal tubule leads to cystogenesis via activation of mammalian/mechanistic target of rapamycin (mTOR) and inactivation of AMP-activated protein kinase (AMPK) activities [3]. Thus, compensatory glucose reabsorption by SGLT1 could increase TKV in the late proximal tubule during dapagliflozin treatment.

A recent study demonstrated that empagliflozin, another SGLT2 inhibitor, increased kidney weight by 13% in wild-type mice as compared to the controls [20]. In addition, hypertrophic or hyperplastic growth of swollen cells in the collecting duct was shown by histological quantification of the tubular cell area. Although details of the mechanisms remain unknown, the authors noted that increased mRNA expression of H₂O-channels (Aqp-2/Aqp-3) induced by an increased level of vasopressin mRNA are likely associated with cell growth in the collecting duct. Furthermore, vasopressin is known as a powerful modulator of cystogenesis based on its effect to increase cAMP level in the collecting duct [21]. Therefore, SGLT2 inhibitor administration may increase TKV via increased vasopressin in ADPKD cases. Additional studies are needed, including examinations of PKD model animals to determine whether hypertrophic or hyperplastic growth associated with cell swelling in the collecting duct is present.

The DAPA–CKD trial, which included subjects with GFR from 25 to 75 mL/1.73 m², noted promising findings indicating attenuated rate of kidney function decline with dapagliflozin treatment, regardless of the presence or absence of diabetes [10]. At the time of initiation of dapagliflozin in the present case, eGFR was 67.3 mL/min/1.73 m², within the same range of criteria used in that trial. However, after three months, eGFR was reduced by 11.1 mL/min/1.73 m² and then by 15.9 mL/min/1.73 m² at 1 year after initiation of dapagliflozin in the present case, which was greater than those values for the participants in the DAPA–CKD trial of approximately 4 and 5 mL/min/1.73 m², respectively [10]. Such discrepancy in eGFR decline rate after the initial dip between that trial and the present case cannot be explained by amelioration of glomerular hyperfiltration, dehydration, or proximal tubular injury, demonstrated by osmotic vacuolization in proximal tubules [22] during dapagliflozin administration. Therefore, increased TKV after dapagliflozin administration may have had a role in that additional eGFR decline in our patient.

Glycosuria elicits an adaptive increase in energy intake, and findings indicating compensatory hyperphagias and

increased energy intake during SGLT2 inhibitor administration have been reported [23, 24]. Since appropriate nutritional diet therapy can prevent an increase in intratubular urinary osmotic pressure and compensatory glucose reabsorption by SGLT1 in the late proximal tubule, nutritional guidance to prevent compensatory hyperphagias in ADPKD patients may be important to suppress an increase in TKV and decrease in eGFR, during SGLT2 inhibitors administration.

Important limitations of this case report include lack of information regarding genotype and changes in body weight during dapagliflozin administration. The effects of SGLT2 inhibitors may vary depending on the status of ADPKD progression risk factors, such as genotype and body weight. Accumulation of further case reports including such information is needed.

In conclusion, this is the first known case of long-term administration of dapagliflozin that showed a continuing increase in TKV and decrease in eGFR in an ADPKD patient in Japan. Although beneficial pleiotropic effects including blood pressure and hemodynamic regulation, lipotoxicity protection, and uric acid control have been found in examinations of SGLT2 inhibitors [25], caution may be needed with ADPKD patients when administering dapagliflozin.

Declarations

Conflict of interest None of the authors have conflicts of interests to declare in regard to dapagliflozin. Regarding SGLT2 inhibitors other than dapagliflozin, KM has received honorariums for lecturing from Mitsubishi Tanabe Pharma Corporation, AstraZeneca K.K., and Nippon Boehringer Ingelheim. KM and ME have received research funding from Tanabe Pharma Corporation.

Ethical approval This article does not contain results of any studies with human participants or animals performed by any of the authors.

Informed consent Written informed consent was obtained from the patient for publication of her clinical data.

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